Art Gallery Problems for Orthogonal Polygons and Orthogonal Polyhedra

A Literature Review of Results and Open Problems

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Abstract

This review surveys art-gallery problems for orthogonal polygons, polyominoes, and orthogonal polyhedra, with emphasis on perimeter-type bounds, holes, and the model changes that produce current open problems. In two dimensions, the perimeter-over-six question is meaningful only after fixing a lattice scale. Maßberg proves the tight $\lfloor \ell/6 \rfloor$ point-guard bound for hole-free polyominoes, and that theorem also covers hole-free integral orthogonal polygons after unit-grid subdivision. The unresolved perimeter-over-six direction is therefore the version with holes, or a broader lattice-domain formulation not reducible to the hole-free polyomino case. In three dimensions, point guards do not provide a direct analogue: orthogonal polyhedra may require $\Theta(n^{3/2})$ point guards. The 3D literature consequently turns to edge guards, reflex-edge guards, face guards, and discrete polycube models, where several Urrutia-type conjectures and gap-closing problems remain open.

Contents

1 Introduction and Scope

The art-gallery literature is best read through four modeling choices: the object class, the presence of holes, the allowed guard model, and the parameter in the bound. A statement about point guards in a hole-free polyomino need not imply a statement about vertex guards in a polygon with holes, and neither statement transfers automatically to edge guards in an orthogonal polyhedron. This review keeps those distinctions explicit because most apparent conflicts among the results come from changing one of these choices.

For two-dimensional orthogonal objects, the main perimeter issue is scale. There is no scale-free theorem of the form "at most $P/6$ guards" for arbitrary real-coordinate orthogonal polygons when P is Euclidean perimeter: scaling the polygon changes P without changing the guard number. Perimeter-over-six results therefore belong to lattice-normalized settings, such as polyomino perimeter, integral perimeter, or the unit-edge ortho-unit model.

Main 2D Distinction

If Q is a hole-free integral orthogonal polygon of lattice perimeter N, unit-grid subdivision identifies it with a hole-free polyomino of perimeter $\ell = N$. Maßberg's theorem then gives $\lfloor N/6 \rfloor$ point guards. Thus the hole-free integral-polygon reading of the Diaz-Banez et al. $\lfloor N/6 \rfloor$ question is already covered by Maßberg; the genuinely unresolved reading is the extension to holes or to a broader lattice-domain convention.

In three dimensions the direct point-guard analogy breaks for a different reason. Orthogonal polyhedra can require superlinear numbers of point guards in the number of vertices, so the 3D branch of the literature usually asks for edge guards, reflex-edge guards, face guards, or guards in discrete polycube visibility models. The word *orthonormal* is not the relevant geometric term; the standard objects are orthogonal polygons, orthogonal polyhedra, and orthogonal polytopes.

The following table summarizes the readings of the perimeter-over-six phrase that recur in the literature.

Reading of the $\lfloor N/6 \rfloor$ question	Status	Reason
Simple/hole-free integral orthogonal polygon	Solved	Same object as a hole-free polyomino after unit-grid subdivision; $N = \ell$.
Ortho-unit polygon	Solved, stronger	Diaz-Banez et al. prove $\lfloor (n-4)/8 \rfloor$, and here $n = P = N$.
Integral or polyomino domain with holes	Open/conditional	Maßberg's hole extension depends on an unproved maximal-rectangle packing conjecture.
Real-coordinate orthogonal polygon with Euclidean perimeter	Not a valid scale-free statement	Uniform scaling changes perimeter but not guard number.

2 Conventions and Terminology

2D objects. An *orthogonal polygon* is a polygon whose edges are horizontal or vertical; the literature also uses *rectilinear* for the same class. A *polyomino* is a finite edge-connected union of unit axis-parallel squares; its perimeter is the number of unit boundary edges, including hole boundaries when holes are present. An *integral orthogonal polygon* has integer coordinates, so its perimeter is measured as a lattice length. An *ortho-unit polygon* is an orthogonal polygon all of whose edges have length one; in that model, the number of vertices equals the perimeter.

3D objects. The standard 3D term is *orthogonal polyhedron*: a polyhedron whose faces are axis-aligned, equivalently whose face normals are coordinate directions. In higher dimensions, the corresponding term is *orthogonal polytope*. The word *orthonormal* describes vectors or bases, not galleries. Polycubes are the voxel analogue of polyominoes and form a discrete subclass of orthogonal polyhedra.

Guard models. Unless a section says otherwise, a 2D guard is a standard point guard with straight-line visibility inside the polygon. In 3D the model must be stated: the literature distinguishes point guards, vertex guards, edge guards, reflex-edge guards, face guards, open edge guards, and discrete polycube visibility graph models. These models are not interchangeable.

Why perimeter needs a lattice scale. No theorem of the form "at most $P/6$ guards" can hold for arbitrary real-coordinate orthogonal polygons if P is Euclidean perimeter: scaling a polygon down preserves the guard number while making P arbitrarily small. The perimeter-over-six question is therefore a lattice-normalized question: polyomino perimeter, integral perimeter, or a fixed unit-edge model.

Common parameters. The same letters are used differently across the literature, so the review uses the following conventions unless a cited theorem uses its own notation.

Symbol	Meaning in this review	Main caution
n	Number of polygon vertices, or polyhedron vertices when the 3D context is explicit.	Vertex-count bounds are not perimeter bounds.
h	Number of holes in a 2D polygonal domain.	Holes can change both decomposition structure and guard type.
r	Number of reflex edges in a 3D orthogonal polyhedron.	This is a 3D parameter; in 2D, reflex vertices play the analogous role.
m	Number of polyomino cells in 2D, or number of polyhedron edges in 3D when stated.	Area/cell bounds and edge-count bounds should not be compared without the model.
ℓ	Lattice perimeter of a polyomino, including hole boundaries.	This is the parameter in Maßberg's perimeter theorem.
N	Lattice perimeter of an integral orthogonal polygon.	After unit-grid subdivision, $N = \ell$ for the corresponding polyomino.
P	Euclidean perimeter when discussing scale, or a polygon only when no formula is involved.	Real-coordinate Euclidean perimeter is not scale-free.

3 Historical Flow

1. Classical art-gallery theorems start with arbitrary simple polygons: Chvatal proves $\lfloor n/3 \rfloor$ and Fisk gives the standard triangulation proof [7, 11]. Shermer, O'Rourke, and Urrutia provide the main survey/book context [26, 22, 27].

2. Orthogonal polygons admit stronger vertex-count theorems: Kahn–Klawe–Kleitman prove the tight $\lfloor n/4 \rfloor$ bound for simple orthogonal polygons [15].

3. Holes complicate the 2D theory. Vertex-guard conjectures for orthogonal polygons with holes, especially Shermer's and Hoffmann's formulations, remain separate from the perimeter-over-six point-guard question [28, 14, 33, 20].

4. Polyominoes introduce area and lattice-perimeter parameters. The tight area theorem allows holes [3], while Maßberg's perimeter-over-six theorem is proven for hole-free polyominoes and becomes conditional in the presence of holes [18, 19].

5. Recent ortho-unit work proves a strong unit-edge theorem and states an integral-perimeter conjecture [8]. Its literal hole-free integral-polygon reading overlaps with Maßberg's polyomino theorem; only the holes or broader-domain reading remains open.

6. In 3D, point guards are too weak for a planar-style theorem: orthogonal polyhedra need $\Theta(n^{3/2})$ point guards in the worst case [23, 29]. Edge-guard and reflex-edge-guard conjectures become the main 3D analogue [2, 31, 6].

4 Status Overview

Table 1 separates the main models discussed below. The labels "proven tight," "conditional," "conjectural," and "open" refer to the exact model in the first three columns; changing the guard type, allowing holes, or changing the parameter can change the answer.

Model	Guards	Parameter	Status	Main source
Simple orthogonal polygon	point	vertices n	Proven tight: $\lfloor n/4 \rfloor$	Kahn–Klawe–Kleitman; Gyori [15, 13]
Orthogonal polygon with holes	vertex	vertices n, holes h	Open at the strongest conjectured bounds; improved h-independent upper bounds and partial sharp cases are known	Urrutia; Hoffmann–Kriegel; Zylinski; Michael–Pinciu [28, 14, 33, 20]
Hole-free polyomino	point	lattice perimeter ℓ	Proven tight for $\ell \geq 6$: $\lfloor \ell/6 \rfloor$	Maßberg [18]
Polyomino with holes	point	lattice perimeter ℓ	Conditional/open: Maßberg's extension depends on a rectangle-packing conjecture	Maßberg habilitation [19]
Polyomino with holes	point	cells m	Proven tight: $\lfloor (m+1)/3 \rfloor$; does not imply perimeter-over-six	Biedl et al. [3]
Ortho-unit polygon	point	perimeter $P = n$	Proven tight for $n \geq 12$: $\lfloor (n-4)/8 \rfloor$, stronger than $P/6$ in this model	Diaz-Banez et al. [8]
Integral orthogonal polygon, hole-free	point	lattice perimeter N	Solved by overlap: subdivision gives a hole-free polyomino with $N = \ell$, so Maßberg proves the Diaz-Banez $\lfloor N/6 \rfloor$ bound in this case	Maßberg; Diaz-Banez et al. [18, 8]
Integral or polyomino domain with holes	point	lattice perimeter N or ℓ	Still open/conditional: Maßberg's hole extension depends on a rectangle-packing conjecture	Maßberg habilitation [19]
Orthogonal polyhedron	point	vertices n	No linear planar analogue: worst case is $\Theta(n^{3/2})$	Paterson–Yao; Viglietta [23, 29]
Orthogonal polyhedron	edge or reflex edge	edges m, reflex edges r	Main 3D analogue; Urrutia-type conjectures remain open beyond partial classes	Benbernou et al.; Viglietta; Cano et al. [2, 31, 6]

Table 1: *Status map for orthogonal art-gallery problems closest to perimeter-over-six.*

Figure 1 summarizes which two-dimensional polygon classes are actually covered by the main bounds in this review. Green cells are proven coverage, pale green cells are proven but usually weaker than a later special theorem, blue cells indicate a proper-subclass or different-parameter theorem, yellow cells mark conditional or open status, and gray cells are outside the theorem as stated.

2D object class	Classical point $\lfloor n/3 \rfloor$	Orthogonal point $\lfloor n/4 \rfloor$	Orthogonal holes, vertex guards	Perimeter point $\lfloor \ell/6 \rfloor$ or $\lfloor N/6 \rfloor$	Area point $\lfloor (m+1)/3 \rfloor$	Ortho-unit point $\lfloor (n-4)/8 \rfloor$
General simple polygon, no holes	yes, tight	–	–	–	–	–
Simple orthogonal polygon, no holes	yes, weaker	yes, tight	–	–	–	–
Hole-free integral orthogonal polygon	yes, weaker	yes, vertex bound	–	yes, by subdivision	yes, area parameter	only unit-edge subclass
Hole-free polyomino	yes, weaker	yes, vertex bound	–	yes, tight	yes, area parameter	only unit-edge subclass
Ortho-unit polygon	yes, weaker	yes, vertex bound	–	yes, follows	yes, area parameter	yes, tight
Orthogonal domain with holes, real coordinates	–	–	partial/open	–	–	–
Integral or polyomino domain with holes	–	–	vertex-only partial/open	cond./open	yes, tight	–

Figure 1: *Two-dimensional coverage chart for the principal bounds. A dash means the theorem does not apply as stated, usually because the object has holes, lacks a lattice scale, uses the wrong guard model, or is controlled by a different parameter.*

5 Model Separations and Non-Implications

The most useful way to test whether a proposed theorem is genuinely new is to ask which model changes are being made. The following separations account for most apparent overlaps in this part of the literature.

Scale separation.
A Euclidean perimeter statement for arbitrary real-coordinate orthogonal polygons cannot be scale-free. Any perimeter-over-six theorem must specify a unit: polyomino boundary length, integral lattice length, or unit-edge ortho-unit length.

Point guards versus vertex guards.
Vertex-guard theorems for orthogonal polygons with holes do not imply point-guard perimeter theorems. Conversely, a point-guard perimeter theorem need not place guards on vertices. This is why Shermer's and Hoffmann's vertex-guard conjectures are adjacent to, but distinct from, the polyomino perimeter question.

Holes versus hole-free domains.
The hole-free perimeter theorem is not automatically stable under adding holes. Holes change the intersection graph of maximal rectangles and break the perfect-graph route used in the hole-free proof. Maßberg's maximal-rectangle packing conjecture is precisely the missing replacement structure in that setting.

Area versus perimeter.
The tight $\lfloor (m+1)/3 \rfloor$ theorem for m-cell polyominoes with holes is an area theorem. It does not imply a perimeter theorem because polyominoes can have very different area-to-perimeter ratios, and the conjectured perimeter bound is sensitive to boundary complexity rather than cell count.

Ortho-unit versus integral.
Ortho-unit polygons form a narrow unit-edge subclass: every edge has length one, so vertex count and perimeter coincide. Integral orthogonal polygons allow longer lattice edges. In the hole-free case, subdivision reduces the integral model to hole-free polyominoes; with holes, it reduces to the polyomino-with-holes case, which is still conditional/open for perimeter.

2D point guards versus 3D point guards.
Orthogonal polyhedra do not inherit planar point-guard behavior. The $\Theta(n^{3/2})$ point-guard bound in 3D forces the analogue to move toward edge guards, reflex-edge guards, face guards, or discrete polycube models.

6 Proof Techniques and Where They Break

Triangulation and coloring.
Chvatal and Fisk establish the classical template for simple polygons [7, 11]. Kahn–Klawe–Kleitman and Gyori adapt coloring ideas to simple orthogonal polygons [15, 13]. The method is powerful for vertex-count bounds but does not directly produce a lattice-perimeter theorem.

Quadrilateralization and cactus duals.
Orthogonal holes are often attacked by decomposing the domain into rectangles or quadrilaterals and coloring a dual structure. Zylinski's cactus-dual condition explains why the $\lfloor (n+h)/4 \rfloor$ conjecture is approachable in some cases but not yet resolved in full [33]. Michael and Pinciu introduce same-sign diagonal graphs of convex quadrangulations and use vertex-cover bounds to unify several orthogonal-guarding results and improve h-independent bounds for polygons with holes [20].

Rectangle packing and perfect-graph methods.

Motwani–Raghunathan–Saran, Worman–Keil, and Maßberg connect orthogonal guarding with star polygons, rectangle systems, and perfect graphs [21, 32, 18]. This is the proof technology behind the hole-free polyomino perimeter theorem. Its failure point is explicit: the hole case depends on Maßberg's maximal-rectangle packing conjecture [19].

Area and grid-cell arguments.

Biedl et al. prove the tight $\lfloor (m+1)/3 \rfloor$ theorem for polyominoes with holes by using the cell structure rather than perimeter [3]. This is strong evidence that holes are tractable in a discrete model, but it leaves open whether perimeter alone controls the guard number.

Hardness and approximation reductions.

Schuchardt–Hecker, Katz–Roisman, and later work on sliding cameras and transmitters show that optimization versions remain hard even in restricted orthogonal settings [25, 16, 9, 4]. These results do not contradict extremal upper bounds; they warn that finding a minimum guard set is a different problem from proving a universal bound.

3D decomposition and binary-space partitions.

Paterson–Yao's binary-space-partition theorem, as used in Viglietta's survey, shows that point guards in orthogonal polyhedra have $\Theta(n^{3/2})$ worst-case behavior [23, 29]. This is the main reason a 3D analogue should not simply ask for a point-guard version of the planar perimeter theorem.

Edge-guard and reflex-edge induction.

Benbernou et al., Viglietta, and Cano–Toth–Urrutia–Viglietta show that the most promising 3D analogue is based on edges, reflex edges, and genus rather than point guards [2, 31, 6]. The remaining gap is structural: the known upper bounds do not meet the Urrutia-type lower-bound constructions.

7 Proof Roadmap for Core Result Lines

This section records what each main proof line actually buys. It is included to make the survey useful for readers who want to enter an open problem rather than only know its status.

7.1 Coloring Proofs in the Plane

The Chvatal–Fisk line proves existence by decomposing a polygon into simple pieces, coloring a graph derived from that decomposition, and choosing the smallest color class [7, 11]. The orthogonal Kahn–Klawe–Kleitman theorem uses the extra parity and degree structure of rectilinear polygons to improve the general $\lfloor n/3 \rfloor$ bound to $\lfloor n/4 \rfloor$ [15]; Gyori's proof is the shortest entry point for this result [13]. The transferable idea is the small-color-class argument. The non-transferable part is the parameter: these proofs count vertices, not lattice perimeter.

7.2 Holes and Quadrilateralization

For orthogonal polygons with holes, the natural decomposition is a quadrilateralization rather than a triangulation. The dual structure can fail to be a tree, so the coloring argument needs additional hypotheses. Zylinski isolates one such hypothesis through cactus-dual quadrilateralizations and recovers Aggarwal's theorem for $h \leq 2$ holes [33]. Michael–Pinciu give a different organizing language: same-sign diagonal graphs and vertex covers [20]. These papers show that holes are not merely a nuisance parameter; they alter the combinatorial object being colored or covered.

This is why the hole literature should be cited carefully. The results are highly relevant to orthogonal domains with holes, but their strongest statements are usually vertex-guard and vertex-count statements. They do not settle point-guard perimeter-over-six.

7.3 Perfect Graphs, Rectangles, and Perimeter

Maßberg's perimeter theorem sits in the rectangle/perfect-graph branch of the literature rather than the triangulation/coloring branch. The proof relates guarding to systems of maximal rectangles, uses perfect-graph structure to control the relevant packing/covering quantity, and then charges that structure to the lattice perimeter [18]. This is why the theorem has the form $\lfloor \ell/6 \rfloor$ and why it is genuinely different from the $\lfloor n/4 \rfloor$ orthogonal-polygon theorem.

The same roadmap also explains the open hole case. In a rectilinear gallery with holes, the intersection graph of maximal rectangles need not retain the same perfect-graph behavior. Maßberg's habilitation identifies a maximal-rectangle packing conjecture as the missing structural substitute [19]. A proof of that conjecture would be a proof route for the perimeter theorem with holes; a counterexample would likely force either a correction term or a different invariant.

7.4 Cell-Count Polyomino Bounds

Biedl et al. prove the tight $\lfloor (m+1)/3 \rfloor$ theorem for m-cell polyominoes, including holes [3]. This proof line uses the cell structure directly, which is why it survives holes cleanly. The result is strong, but it answers a different extremal question: how guard number scales with area, not with boundary complexity. For research purposes, the important lesson is that holes are not inherently fatal in polyomino models; they are fatal to the specific perimeter proof technology currently known.

7.5 3D Decomposition and Guard Models

In three dimensions, the first modeling decision is guard type. The Paterson–Yao/Viglietta point-guard line shows that orthogonal polyhedra can require $\Theta(n^{3/2})$ point guards [23, 29]. That fact rules out a direct planar-style point-guard theorem in terms of vertices or edges.

The promising 3D analogy therefore uses edges. Benbernou et al. improve orthogonal-polyhedron edge-guard and reflex-edge upper bounds [2]; Viglietta proves tight-style results for 2-reflex orthogonal polyhedra [31]; and Cano–Toth–Urrutia–Viglietta improve the general polyhedron edge-guard bound [6]. The research gap is not whether edge guards are relevant, but whether the existing decomposition and charging methods can be sharpened to the lower-bound scale.

8 Status by Model

Simple orthogonal polygons, point guards.

Proven tight: every n-vertex simple orthogonal polygon is guardable by $\lfloor n/4 \rfloor$ point guards [15, 13].

Orthogonal polygons with holes, vertex guards.

Several upper bounds and partial sharp results are known. O'Rourke gives the classical $\lfloor (n+2h)/4 \rfloor$ style bound for h holes [22]. Shermer's stronger $\lfloor (n+h)/4 \rfloor$ conjecture is open in general; Zylinski gives a coloring proof of Aggarwal's theorem for $h \le 2$ and cactus-dual quadrilateralizations [28, 33]. In the h-independent direction, Hoffmann and Kriegel prove an $n/3$ upper bound, and Michael–Pinciu improve this line to $\lfloor (17n-8)/52 \rfloor$ using diagonal graphs and vertex covers [14, 20].

Hole-free polyominoes, point guards, lattice perimeter ℓ.

Proven tight: for $\ell \ge 6$, Maßberg proves that $\lfloor \ell/6 \rfloor$ point guards suffice [18]. Use $\max\{1, \lfloor \ell/6 \rfloor\}$ if the one-square case is included.

Polyominoes with holes, point guards, lattice perimeter ℓ.

Open/conditional. Maßberg's text explicitly states that the extension to holes would follow from a maximal-rectangle packing conjecture [19].

Polyominoes with holes, point guards, area m.

Proven tight: every m-cell polyomino, holes allowed, is guardable by $\lfloor (m+1)/3 \rfloor$ point guards [3]. This does not settle perimeter-over-six.

Ortho-unit polygons, point guards.

Proven tight: every ortho-unit polygon with $n \geq 12$ vertices is guardable by $\lfloor (n-4)/8 \rfloor$ guards [8]. Since $n = P$ in the ortho-unit model, this is stronger than $P/6$ for that narrow class.

Integral orthogonal polygons, point guards, perimeter N.

Hole-free case proven via Maßberg: after subdividing long boundary edges into unit grid edges, a hole-free integral orthogonal polygon is a hole-free polyomino with $N = \ell$. Diaz-Banez et al. state the natural $N/6$ conjecture in the broader integral-perimeter discussion [8]; its unresolved content is therefore beyond the hole-free case, notably holes.

Polycubes and polyhypercubes.

Pinciu studies discrete polyform variants, including polycubes and polyhypercubes [24]. These are relevant for 3D lattice analogues, but they use a discrete polyform/visibility-graph framework rather than the continuous point-guard model for orthogonal polyhedra.

Orthogonal polyhedra, point guards.

No planar-style linear theorem exists in terms of vertices. The Paterson–Yao bound gives $\Theta(n^{3/2})$ point guards as sufficient and sometimes necessary for n-vertex orthogonal polyhedra [23, 29].

Orthogonal polyhedra, edge and reflex-edge guards.

The central open direction is an Urrutia-type edge-guard theorem: lower-bound examples require roughly $m/12$ edge guards, while known upper bounds are larger. Benbernou et al. improve the orthogonal upper bound to asymptotic $(11/72)m$ edge guards, or $(7/12)r$ in terms of reflex edges [2]. Viglietta proves optimal-style bounds for 2-reflex orthogonal polyhedra [31]. Cano, Toth, Urrutia, and Viglietta improve the general 3D polyhedron edge-guard bound to $(5/6)m$, while noting a substantial remaining gap [6].

9 Two-Dimensional Orthogonal Polygons

9.1 Classical Background

The original art-gallery theorem is due to Chvatal: $\lfloor n/3 \rfloor$ guards always suffice for a simple n-vertex polygon, and this is tight [7]. Fisk's short proof via triangulation and three-coloring became the standard proof template [11]. O'Rourke's book remains the standard comprehensive reference for the classical theory [22]; Shermer and Urrutia survey art-gallery and illumination problems, including variants and open problems [26, 27]. On the algorithmic side, Abrahamsen, Adamaszek, and Miltzow prove that the general art-gallery problem is $\exists\mathbb{R}$-complete even for simple polygons with integer-coordinate vertices, underscoring why restricted orthogonal and lattice models are usually treated separately [1, Theorem 1].

9.2 Simple Orthogonal Polygons

Kahn, Klawe, and Kleitman prove the tight orthogonal theorem: $\lfloor n/4 \rfloor$ point guards suffice for every simple orthogonal polygon with n vertices [15]. Gyori gives a short proof [13]. This theorem is a vertex-count theorem, not a perimeter theorem. In a lattice setting the vertex count and the perimeter may be very different, which is why the polyomino and integral-perimeter questions are separate.

The proof tradition connects to decomposition, coloring, and perfect-graph methods. Motwani, Raghunathan, and Saran study covering orthogonal polygons with star polygons [21]; Worman and Keil develop decomposition and visibility-graph machinery for orthogonal art-gallery problems [32]. Bjørling-Sachs proves the tight $\lfloor n/4 \rfloor$ edge-guard bound for rectilinear polygons, a useful 2D reference point for later edge-guard models [5].

9.3 Holes

Holes are one of the main fault lines in the literature. For orthogonal polygons with h holes, classical vertex-guard bounds exist, but the strongest natural conjectures are not settled in general. Urrutia's open-problem list records Shermer's conjecture that an n-vertex orthogonal polygon with h holes should be guardable by $\lfloor (n+h)/4 \rfloor$ vertex guards, and Hoffmann's conjecture that $\lfloor 2n/7 \rfloor$ vertex guards should suffice for orthogonal polygons with holes [28]. Zylinski proves the $\lfloor (n+h)/4 \rfloor$ bound under a cactus-dual quadrilateralization condition and recovers Aggarwal's theorem for $h \leq 2$ [33]. Hoffmann and Kriegel prove graph-coloring consequences for rectilinear polygons with holes, including upper bounds for rectilinear prison-yard variants [14]. Michael and Pinciu later reframe much of this orthogonal vertex-guard theory through same-sign diagonal graphs and vertex covers; in particular, they give an h-independent upper bound $\lfloor (17n-8)/52 \rfloor$ for orthogonal polygons with any number of holes [20].

These hole results are relevant context, but they do not prove the perimeter-over-six statement. They usually concern vertex guards and vertex counts; the perimeter-over-six problem concerns point guards and lattice perimeter.

10 Polyomino and Perimeter Results

10.1 Hole-Free Polyominoes

Maßberg proves the direct perimeter theorem for hole-free polyominoes [18]. In the notation of that paper, if a hole-free polyomino has perimeter $\ell \geq 6$, then it can be guarded by at most $\lfloor \ell/6 \rfloor$ point guards. The proof uses maximal axis-parallel rectangles in a rectilinear gallery, relates a rectangle packing number to the guard number, and then bounds the packing by the perimeter. Comb-like examples show tightness.

10.2 Polyominoes with Holes

The area theorem does allow holes: Biedl, Irfan, Iwerks, Kim, and Mitchell prove that every m-cell polyomino can be guarded by $\lfloor (m+1)/3 \rfloor$ point guards, and the bound is tight [3]. This is the strongest clean theorem for arbitrary polyominoes with holes in the standard point-guard model, but it is an area bound rather than a perimeter bound.

For perimeter, the hole case remains conditional. Maßberg observes that the perimeter theorem would extend to holes if a maximal-rectangle packing conjecture were true [19]. This is *not* a proof of perimeter-over-six for holes; it identifies the missing structural statement.

10.3 Ortho-Unit and Integral Orthogonal Polygons

Diaz-Banez et al. prove that every ortho-unit polygon with $n \geq 12$ vertices can be guarded by $\lfloor (n-4)/8 \rfloor$ guards, and that the bound is tight [8]. Since every edge has unit length in an ortho-unit polygon, n is the perimeter. This solves a narrow unit-edge class more strongly than $P/6$.

The same paper frames the integral-perimeter conjecture: every integral orthogonal polygon of perimeter N should be guardable by $\lfloor N/6 \rfloor$ point guards. If "integral orthogonal polygon" is read in the usual simple, hole-free sense, this is already covered by Maßberg. Subdividing long sides into unit grid edges turns the polygon into a hole-free polyomino with the same lattice perimeter, $N = \ell$, and Maßberg proves the $\lfloor \ell/6 \rfloor$ bound for that class. Thus the Diaz-Banez question is open only under a broader reading, such as integral domains with holes or another lattice-domain convention not reduced to a hole-free polyomino.

11 Three-Dimensional Orthogonal Polyhedra

11.1 Point Guards

The direct 3D point-guard analogue of planar art-gallery theorems fails. In orthogonal polyhedra with n vertices, Paterson and Yao's binary-space partition result implies a tight $\Theta(n^{3/2})$ point-guard bound [23]. Viglietta's thesis presents this as Theorem 3.7 and draws the key modeling conclusion: point guards do not yield a linear orthogonal-polyhedron art-gallery theorem [29]. Therefore a 3D "perimeter-over-six" theorem is not obtained by simply replacing polygons with orthogonal polyhedra and keeping point guards.

11.2 Edge and Reflex-Edge Guards

The 3D literature often treats edge guards as the analogue of planar vertex guards. Urrutia conjectured that an orthogonal polyhedron with m edges should be guardable by $m/12 + O(1)$ closed edge guards, matching known lower bound examples up to an additive constant [29]. A related reflex-edge conjecture asks whether roughly $(r - g)/2 + 1$ reflex edge guards suffice, where r is the number of reflex edges and g is the genus.

Benbernou, Demaine, Demaine, Kurdia, O'Rourke, Toussaint, Urrutia, and Viglietta improve the known asymptotic upper bounds for orthogonal polyhedra: $(11/72)m$ edge guards suffice in terms of total edges, and $(7/12)r$ suffice in terms of reflex edges [2]. Viglietta later proves optimal-style bounds for 2-reflex orthogonal polyhedra, namely orthogonal polyhedra whose reflex edges occur in only two directions [31]. These are major partial results, but they do not settle the general 3-reflex orthogonal-polyhedron conjectures.

Cano, Toth, Urrutia, and Viglietta give a broad theorem for arbitrary polyhedra in three-space: every polyhedron with m edges can be guarded by at most $(5/6)m$ edge guards [6]. This improves the general-polyhedron edge-guard bound, but it still leaves a large gap from the conjectured $m/6$ style lower/upper behavior for general polyhedra and the $m/12 + O(1)$ orthogonal-polyhedron conjecture.

11.3 Face Guards and Polycube Variants

Face guards are another 3D model. Viglietta studies face-guarding for several classes of polyhedra, including orthogonal polyhedra, 4-oriented polyhedra, and 2-reflex orthostacks [30]. Face guards are useful for mapping the 3D model space, but they are not a direct continuation of the point-guard perimeter-over-six question.

For lattice 3D objects, Pinciu's work on polycubes and polyhypercubes gives a discrete polyform line of results [24]. This is the natural place to look when the 3D model is voxel/discrete rather than continuous orthogonal-polyhedron visibility.

12 Algorithmic and Optimization Context

Extremal art-gallery theorems answer a worst-case existence question: how many guards always suffice for every object in a class? Optimization papers ask a different question: given one object, how hard is it to find a minimum guard set? The distinction matters because a clean universal upper bound need not yield an efficient exact algorithm for the minimum set, and an NP-hardness result does not rule out a sharp extremal theorem.

Lee and Lin established the classical NP-hardness framework for art-gallery optimization problems [17]. In orthogonal settings, Schuchardt–Hecker prove NP-hardness for two orthogonal art-gallery problems [25], and Katz–Roisman study hardness for guarding vertices of rectilinear domains [16]. At the level of the general point-guard problem, Abrahamsen–Adamaszek–Miltzow show $\exists\mathbb{R}$-completeness, already for simple polygons with integer-coordinate vertices [1, Theorem 1]. These complexity results should be read as optimization barriers, not as barriers to universal counting theorems such as $\lfloor n/4 \rfloor$ or $\lfloor \ell/6 \rfloor$.

Approximation and restricted-visibility models form a parallel literature. Ghosh gives broad approximation-algorithm context for art-gallery problems [12]. Durocher et al. and Biedl et al. study sliding cameras and sliding k-transmitters in orthogonal polygons, including holes and hardness or approximation results in those models [9, 4]. Filtser et al. study polyominoes under k-hop visibility, a discrete variant close to network or city-block applications rather than straight-line point-guard visibility [10]. These variants are useful sources

Problem cluster	Status source	What is known	What would constitute progress
Polyomino perimeter with holes	Maßberg habilitation [19]	Hole-free theorem is published; hole extension is conditional on a maximal-rectangle packing conjecture.	Prove the packing conjecture, prove $\lfloor \ell/6 \rfloor$ by another method, or construct a hole counterexample.
Integral perimeter beyond the hole-free case	Diaz-Banez et al.; Maßberg [8, 18]	Hole-free integral polygons reduce to hole-free polyominoes.	State and resolve the exact holes/broader-domain version, including how perimeter is counted.
Orthogonal polygons with holes, vertex guards	Urrutia; Zylinski; Michael–Pinciu [28, 33, 20]	Strong conjectures remain open; partial and h-independent upper bounds are known.	Close the gap between lower-bound constructions and the best vertex-guard upper bounds.
3D orthogonal edge guards	Viglietta; Benbernou et al. [29, 2]	Known upper bounds exceed Urrutia-type lower-bound targets.	Prove an $m/12 + O(1)$ style theorem or identify a larger necessary correction term.
3D reflex-edge guards	Viglietta [31]	2-reflex orthogonal polyhedra are understood much better than general 3-reflex orthogonal polyhedra.	Extend the induction/charging methods beyond the 2-reflex restriction.
Algorithmic restricted cases	Lee–Lin; Schuchardt–Hecker; Ghosh [17, 25, 12]	Exact optimization is hard in many settings; approximation and special visibility models remain active.	Find tractable subclasses aligned with the extremal perimeter or edge-guard conjectures.

Table 2: *Research-facing status table for the main open-problem clusters.*

of techniques and open problems, but they should not be cited as settling the standard perimeter-over-six question.

13 Open Problems and Research Directions

The following problems are the most directly connected to the survey above. Items are grouped by mathematical model rather than by perceived difficulty. When a question is only a consequence of comparing known results, the text says so explicitly; otherwise the cited source records the conjecture or the conditional dependency.

1. **Perimeter-over-six for polyominoes with holes.** Let P be a polyomino with holes and total lattice perimeter ℓ, counting both outer and hole boundaries. Is P always guardable by $\max\{1, \lfloor \ell/6 \rfloor\}$ point guards? Maßberg's habilitation gives a conditional route to this theorem through a maximal-rectangle packing conjecture [19]; the published perimeter theorem is the hole-free case [18].

2. **Maßberg's maximal-rectangle packing conjecture.** In a rectilinear gallery with holes, does the maximum packing size of maximal rectangles control the required number of point guards? This is the structural conjecture identified by Maßberg as sufficient for extending the perimeter-over-six theorem to polyominoes with holes [19].

3. **Integral-perimeter bounds beyond the hole-free case.** The hole-free integral orthogonal polygon version follows from Maßberg by unit-grid subdivision. The remaining integral-perimeter question is whether the same $\lfloor N/6 \rfloor$ behavior survives when holes or broader lattice-domain conventions are allowed [18, 8].

4. **Vertex guards for orthogonal polygons with holes.** Shermer's conjecture asks whether every n-vertex

orthogonal polygon with h holes is guardable by $\lfloor (n+h)/4 \rfloor$ vertex guards. Hoffmann's conjecture asks whether $\lfloor 2n/7 \rfloor$ vertex guards always suffice. These are source-stated open problems in the orthogonal-with-holes literature, and they are related to but distinct from perimeter-type point-guard bounds [28, 14, 33, 20].

5. **A correction term if perimeter-over-six fails with holes.** This is a derived direction rather than a named conjecture. If holes force counterexamples to a pure $\ell/6$ theorem, a useful replacement may need to account for total perimeter, outer perimeter, number of holes, hole perimeter, reflex vertices, or rectangle-packing obstructions.

6. **Urrutia-type edge-guard bounds for orthogonal polyhedra.** For genus-zero orthogonal polyhedra with m edges, the central conjectural target is an $m/12 + O(1)$ edge-guard bound. Lower-bound examples and current upper bounds leave a substantial gap [29, 2].

7. **Reflex-edge guards in general orthogonal polyhedra.** Viglietta proves tight-style bounds for 2-reflex orthogonal polyhedra, where reflex edges occur in only two coordinate directions, but the analogous reflex-edge conjecture for general orthogonal polyhedra remains open [31].

8. **The general three-dimensional edge-guard gap.** Cano, Toth, Urrutia, and Viglietta improve the edge-guard upper bound for arbitrary polyhedra to $(5/6)m$, but explicitly leave the gap between known lower and upper bounds as an open problem [6]. Orthogonal polyhedra form the more structured subclass where sharper Urrutia-type conjectures are expected.

9. **Algorithmic complexity and approximation in restricted models.** Extremal guard bounds and optimization algorithms remain separate questions. Minimum guarding is NP-hard in many orthogonal settings [17, 25, 16, 3, 9, 4]. A parallel algorithmic direction is to identify exact polynomial subclasses and approximation guarantees for polyominoes with holes, integral orthogonal domains, polycubes, and orthogonal polyhedra. Recent work on k-hop visibility in polyominoes records related approximation questions [10].

14 Annotated Source Guide

Foundational Art-Gallery Theory

Chvatal [7] and Fisk [11].

Foundational $\lfloor n/3 \rfloor$ theorem for simple polygons and the short triangulation/coloring proof.

O'Rourke [22].

Standard book reference for art-gallery theorems, algorithms, visibility, decompositions, and early 3D context.

Shermer [26] and Urrutia [27].

Survey references for art-gallery and illumination problems, useful for terminology, variants, and classical open-problem context.

Lee–Lin [17].

Classical computational-complexity source for art-gallery optimization problems.

Abrahamsen–Adamaszek–Miltzow [1].

Modern complexity reference proving $\exists\mathbb{R}$-completeness for the general art-gallery problem, even for simple integer-coordinate polygons.

2D Orthogonal Polygons and Holes

Kahn–Klawe–Kleitman [15].
Primary source for the tight $\lfloor n/4 \rfloor$ theorem for simple orthogonal polygons.

Gyori [13].
Short proof of the rectilinear art-gallery theorem.

Hoffmann–Kriegel [14].
Graph-coloring result with consequences for rectilinear polygons with holes and prison-yard variants.

Zylinski [33].
Coloring proof of Aggarwal's theorem for orthogonal galleries with holes, including the $\lfloor (n+h)/4 \rfloor$ bound for $h \leq 2$ and cactus-dual cases.

Michael–Pinciu [20].
Same-sign diagonal graph and vertex-cover framework for orthogonal polygons, including h-independent bounds for polygons with holes.

Urrutia open-problem page [28].
Direct source for Shermer's and Hoffmann's open conjectures on orthogonal polygons with holes using vertex guards.

Schuchardt–Hecker [25].
NP-hardness source for art-gallery problems in orthogonal polygons.

Worman–Keil [32].
Decomposition and visibility-graph machinery for the orthogonal art-gallery problem.

Bjørling-Sachs [5].
Tight edge-guard theorem for rectilinear polygons; relevant when comparing 2D orthogonal edge guards with 3D edge-guard conjectures.

Katz–Roisman [16].
NP-hardness for guarding vertices of rectilinear domains by vertex, boundary, or point guards.

Polyomino, Perimeter, and Lattice Variants

Maßberg [18].
Primary peer-reviewed source for perimeter-over-six in hole-free polyominoes. This also proves the hole-free integral orthogonal polygon $\lfloor N/6 \rfloor$ statement after unit-grid subdivision.

Maßberg habilitation [19].
Source for Conjecture 6.10 and the conditional hole extension. This is the key source for the conditional status of the perimeter theorem with holes.

Biedl–Irfan–Iwerks–Kim–Mitchell [3].
Tight $\lfloor (m+1)/3 \rfloor$ area theorem for polyominoes, holes allowed, plus complexity context.

Diaz-Banez et al. [8].
Recent tight theorem for ortho-unit polygons and source of the integral orthogonal polygon $\lfloor N/6 \rfloor$ conjecture. Its simple/hole-free reading overlaps with Maßberg rather than creating a separate open problem.

Pinciu [24].
Polyform extension including polyominoes, polycubes, and polyhypercubes. Relevant for discrete 3D analogues.

Filtser–Krohn–Nilsson–Rieck–Schmidt [10].
Current work on k-hop visibility in polyominoes, including open approximation questions for polyominoes with holes.

3D Orthogonal Polyhedra

Paterson–Yao [23].
Binary-space partition result used for the $\Theta(n^{3/2})$ point-guard bound in orthogonal polyhedra.

Viglietta thesis [29].
Central survey and source for guarding/searching polyhedra, the point-guard obstruction, and Urrutia-type edge-guard conjectures.

Benbernou et al. [2].
CCCG source for improved edge-guard bounds in orthogonal polyhedra: asymptotic $(11/72)m$ and $(7/12)r$.

Viglietta 2-reflex paper [31].
Proves optimal-style reflex-edge bounds for 2-reflex orthogonal polyhedra and gives an $O(n \log n)$ guard-placement algorithm.

Viglietta face-guarding paper [30].
Face-guarding results for several classes of polyhedra, including orthogonal subclasses.

Cano–Toth–Urrutia–Viglietta [6].
General edge-guard result in three-space and explicit open gap.

Algorithms and Related Visibility Models

Motwani–Raghunathan–Saran [21].
Perfect-graph approach to covering orthogonal polygons with star polygons.

Ghosh [12].
General approximation-algorithm context for art-gallery problems.

Durocher et al. [9].
Sliding-camera model for orthogonal art galleries, including holes.

Biedl et al. [4].
Sliding k-transmitter hardness and approximation results for orthogonal polygons.

15 Evidence Audit Trail

The review is backed by a structured evidence layer in the repository. The source registry records bibliographic information and links; the claims registry records the specific theorem, conjecture, hardness, survey, or open-problem statement extracted from each source; the coverage matrix records which model cells are proved, open, conditional, synthesized, or still in need of follow-up; and the open-problems ledger records the unresolved questions and their supporting sources.

The generated evidence report is intended as an audit companion to the prose. It lists central proved results, open and conditional problems, coverage cells, missing page or theorem locators, and next actions for improving the review. The prose above should be updated only after the corresponding source, claim, coverage, and open-problem records support the statement being added.

16 Conclusion

1. The theorem "$\ell/6$ guards suffice" is established for *hole-free polyominoes*. The polyomino-with-holes version is conditional on Maßberg's rectangle-packing conjecture in the sources surveyed.

2. The statement "an integral orthogonal polygon of perimeter N is guardable by $\lfloor N/6 \rfloor$ guards" has two different readings. The simple/hole-free reading is already proved by Maßberg, because it is exactly the hole-free polyomino case after unit-grid subdivision. The holes or broader-domain reading remains open/conditional.

3. The right 3D terminology is *orthogonal polyhedron* or *orthogonal polytope*. The natural 3D guard models are not obtained by copying the 2D point-guard setting.

4. A 3D point-guard analogue of the planar perimeter theorem is blocked by the $\Theta(n^{3/2})$ point-guard bound. The central 3D problems instead concern edge guards, reflex-edge guards, face guards, or discrete polycube visibility models.

5. The most direct open directions are the polyomino-with-holes perimeter bound, Maßberg's rectangle-packing conjecture, the holes/broader-domain $\lfloor N/6 \rfloor$ extension, and the Urrutia-type $m/12 + O(1)$ and reflex-edge conjectures for general orthogonal polyhedra.

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